Irvin Chadraa

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Education

New York University - Courant Institute of Mathematical Sciences

Master of Science in Mathematics - GPA: 3.89 *May 2025*

Bachelor of Science in Mathematics & Minor in Computer Science - GPA: 3.63 *May 2023*

- Founder's Day Award, University Honors Scholar, Dean's List (All Semesters)

Research Interests: *World Models, Reinforcement Learning, Robot Intelligence, Optimization, Computer Vision*

Technical Skills

Programming: Python, C/C++, MATLAB, SQL/MySQL, Java, Scala, Linux
Machine Learning: PyTorch, TensorFlow, Scikit-Learn, OpenCV, ROS, HuggingFace
Big Data/Cloud Technologies: Apache Hadoop, Apache Spark, Git, Jira

Experience

NYU Computational Intelligence, Learning, Vision, and Robotics Lab *May 2025 - Present*

Visiting Researcher

- Investigating zero-shot robot planning for long-horizon goal-reaching tasks using learned world models supported with pre-trained visual encoders.
- Using goal-conditioned behavior cloning policies to warm-start model-predictive planning for robot manipulation tasks to improve sample efficiency and runtime.
- Optimizing planned latent actions through importance sampling and gradient-based methods

NYU Tandon School of Engineering *Sep 2023 - Present*

Graduate Researcher

- Led a team of graduate students to implement a scalable RL framework for autonomous trading, applying deep recurrent architecture and data preprocessing via PCA and discrete wavelet transform
- Creating a benchmark suite of financial RL environments, trading algorithms, and industry-standard baselines.

NYU Center for Urban Science and Progress *June 2023 - Sep 2025*

Graduate Researcher - funded by Office of Naval Research

- Developed a game-theoretic optimal transport algorithm in Python and MATLAB for selectively registering a subset of informative features in high-dimensional remote sensing data, including images and LiDAR point clouds.
- Presented work at *20th 3D GeoInfo / 9th Smart Data Smart Cities Conference* in Japan.

Isomer Space Program *Feb 2022 - June 2023*

Flight Software Engineer

- Co-founded a student-led nonprofit focused on developing a nanosatellite to detect cosmic radiation in low-Earth orbit after successfully raising over $30k in crowdsourced funds.
- Developed flight software; programmed subtasks and built UART/I2C communication pipeline between custom on-board microcontroller and multi-sensor payload, including 16-bit ADC, IMU, and UHF radio transceiver
- Ran hardware-in-the-loop tests before handoff to SpaceX for integration and launch.

iRobot *Jun 2021 - Dec 2021*

Machine Learning Intern

- Optimized CNN-based object detection model for Roomba j7+ with TensorFlow, achieving 20% accuracy gains and contributing to the success of iRobot's top-selling product.
- Implemented computer vision pipelines leveraging OpenCV for data annotation, feature extraction, and high-quality ground truth data generation, leading to a 15% improvement in obstacle avoidance.

Publications

Partial Optimal Transport for Co-Registration of Partially-Overlapped Point Clouds

- **Chadraa, I.**, Tabak, E., & Laefer, D.F. (2025)
- Published in *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences.*

Teaching Experience

NYU Tandon School of Engineering Fall 2024 - Spring 2025

Graduate Teaching Assistant

- VIP-GY 5000: ML and Time-Series Forecasting for Active Portfolio Management

NYU Courant Institute of Mathematical Sciences Fall 2023 - Fall 2024

Graduate Teaching Assistant

- MA-UY 4414: Applied Partial Differential Equations (Fall 2023)
- MA-UY 4434: Applied Complex Variables (Spring 2024)
- MA-UY 4444: Intro to Math Modeling (Fall 2023, Fall 2024)

Boston College Morrissey School of Arts and Science Fall 2020

Math Department Grader

- MATH-1100: Calculus I
- MATH-1101: Calculus II

Projects

LLMs with Quaternion Networks | *Python, PyTorch, BERT, HuggingFace*

- Researched and developed quaternion-augmented transformer models for sentiment analysis and zero-shot time series forecasting, achieving 75% gain in memory efficiency over traditional large language models.

Chatbot for Financial Data | *Python, LLaMA-2-GPTQ, LangChain, Hugging Face*

- Developed a chatbot using a retrieval-augmented generation (RAG) pipeline to forecast stock performance and stream insights of S&P 500 companies via stock prices, SEC filings, & Glassdoor reviews.

Quality of Life in NYC Boroughs | *Spark, Scala, Java, Hadoop, Tableau*

- Utilized Scala in Spark client shell on HDFS via NYU's HPC platform for ETL processes analyzing large volume datasets on property value, EMS dispatch frequency, and water quality from NYC OpenData.